

Number Game (subset)

Rich Uncle Pennybags is getting tired of playing Monopoly, hence developed a new exciting *number game*. In this game, V tokens are randomly extracted, each depicting a distinct positive integer. Each player then has to select a subset S such that every pair (S_i, S_j) of elements in this subset satisfies: $S_i \% S_j = 0$ or $S_j \% S_i = 0$. The player who picks up the bigger subset wins!



Rich Uncle Pennybags is not used to lose a game: help him find the largest allowed subset!

🔗 Among the attachments of this task you may find a template file `subset.*` with a sample incomplete implementation.

Input

The first line contains the only integer N indicating the cardinality (number of elements) of the set V . The following line contains N integers, the elements V_i of the set.

Output

The output should be a single integer, indicating the *largest cardinality* of a subset S of V such that the condition is satisfied. If there are multiple solution, output any of them.

Constraints

- $1 \leq N \leq 1000$.
- $V_i < 2^{32}$ for any i .

Scoring

Your program will be tested against several test cases, and your score will proportional to the number of correctly solved test cases.

Examples

stdin/input.txt	stdout/output.txt
5 1 2 4 8 5	4